Optimizing Molecular Dynamics

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- **Intranode optimization: CPU & memory access**
- **Internode optimization: Communication**

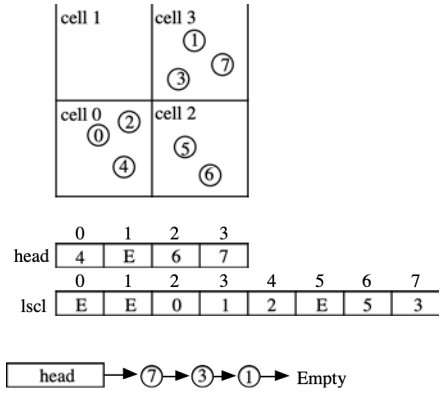
Data/computation locality!



Intranode: Memory Access

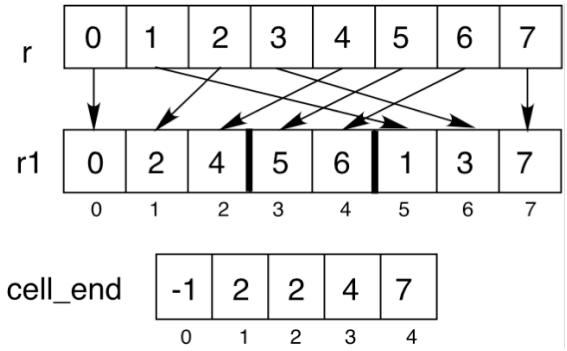
Data re-ordering

- **Linked-list cells—irregular memory access pattern**



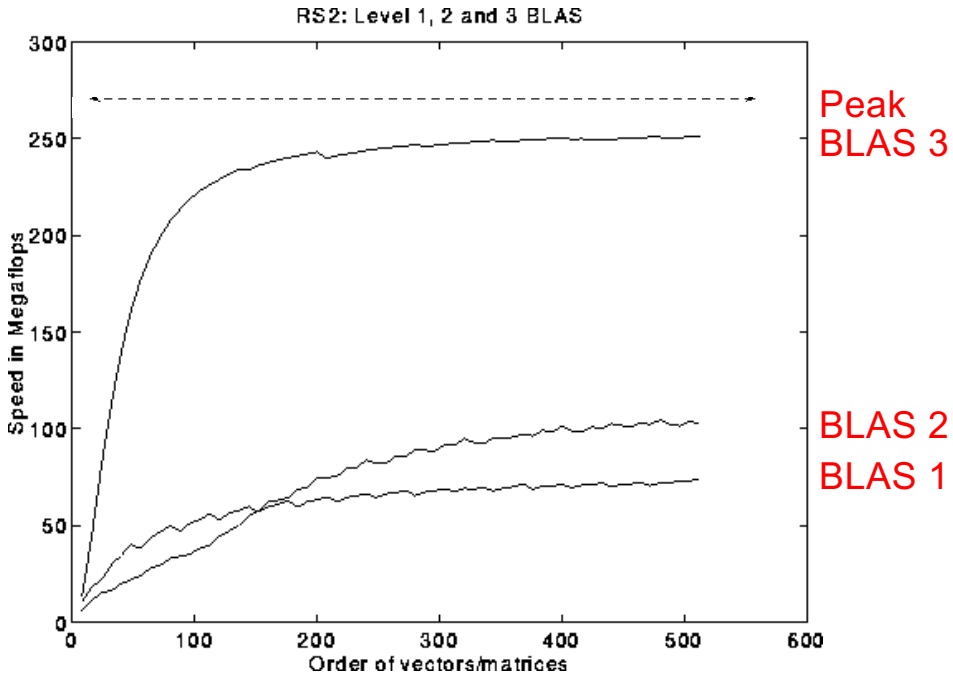
- **Data locality: Regular data layout**

```
for i = cell_end[c]+1 to cell_end[c+1]
  access r1[i]
endfor
```



BLAS3-Performance Molecular Dynamics?

- **BLAS3: q = flop/memory access = (block size)$^{1/2}$**



- **Molecular dynamics: $q = O(n^2)/O(n) = O(n$: block size)**
  - **> Use of SIMD (single instruction multiple data) instructions on multicore (AVX), GPU**

BLAS Floating-Point Performance

- **BLAS-ification: Transform from band-by-band to all-band computations to utilize a matrix-matrix subroutine (DGEMM) in the BLAS3 library for the quantum molecular dynamics application**
- **Algebraic transformation of computations**

Example: Nonlocal pseudopotential operation

D. Vanderbilt, *Phys. Rev. B* **41**, 7892 ('90)

$$\hat{v}_{\mathrm{nl}}|\psi_n^\alpha\rangle = \sum_I^{N_{\mathrm{atom}}} \sum_{ij}^{L_{\mathrm{max}}} |\beta_{i,I}\rangle D_{ij,I} \langle\beta_{j,I}|\psi_n^\alpha\rangle \quad (n = 1, \ldots, N_{\mathrm{band}})$$

$$\boldsymbol{\Psi} = \left[|\psi_1^\alpha\rangle, \ldots, |\psi_{N_{\mathrm{band}}}^\alpha\rangle\right] \quad \widetilde{\mathbf{B}}(i) = \left[|\beta_{i,1}\rangle, \ldots, |\beta_{i,N_{\mathrm{atom}}}\rangle\right] \quad \left[\widetilde{\mathbf{D}}(i,j)\right]_{I,J} = D_{ij,I}\delta_{IJ}$$

$$\hat{v}_{\mathrm{nl}}\boldsymbol{\Psi} = \sum_{i,j}^{L} \widetilde{\mathbf{B}}(i)\widetilde{\mathbf{D}}(i,j)\widetilde{\mathbf{B}}(j)^T$$

- **50.5% of the theoretical peak FLOP/s performance on 786,432 Blue Gene/Q cores (entire Mira at the Argonne Leadership Computing Facility)**
- **55% of the theoretical peak FLOP/s on Intel Xeon E5-2665**

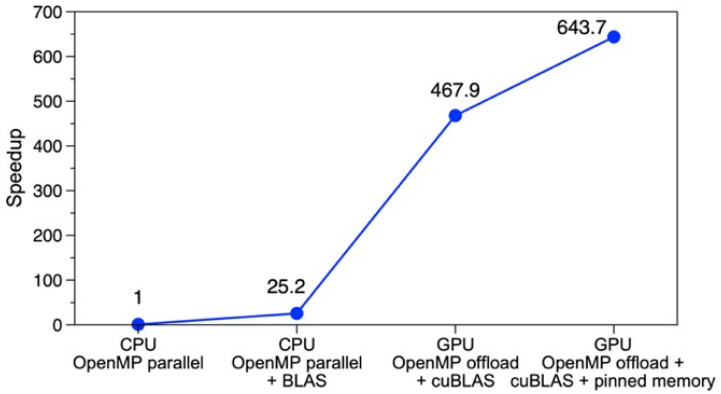
K. Nomura *et al.*, *IEEE/ACM Supercomputing, SC14* ('14)

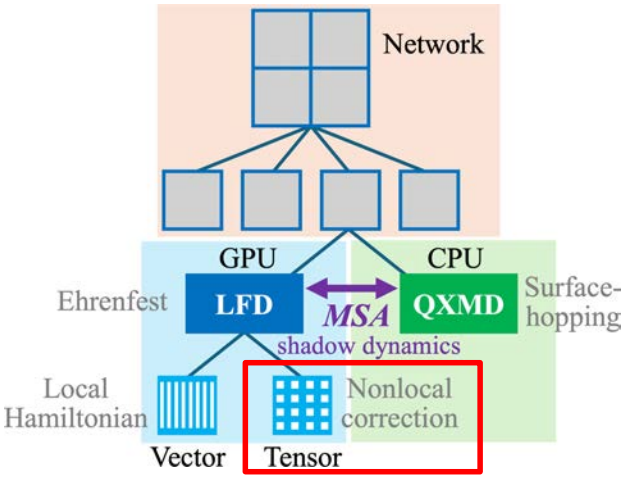
More BLAS-ification

- **Switch of representation from finite-difference to electronic orbital bases makes nonlocal correction into dense complex-matrix multiplications**

$$\underbrace{\widetilde{\Psi(t)}^{N_{\mathrm{grid}}\times N_{\mathrm{orb}}}}_{\text{FP32}} \mathrel{-}= \delta \underbrace{\widetilde{\Psi(0)}^{N_{\mathrm{grid}}\times N_{\mathrm{orb}}}\ \widetilde{\Psi^{\dagger}(0)}^{N_{\mathrm{orb}}\times N_{\mathrm{grid}}}\ \widetilde{\Psi(t)}^{N_{\mathrm{grid}}\times N_{\mathrm{orb}}}}_{\text{BF16}}$$

- **FP32 GEMM emulation with BF16**





Achieved 1.87 Exaflop/s on 60,000 GPUs of Aurora

2025 Gordon Bell prize finalist

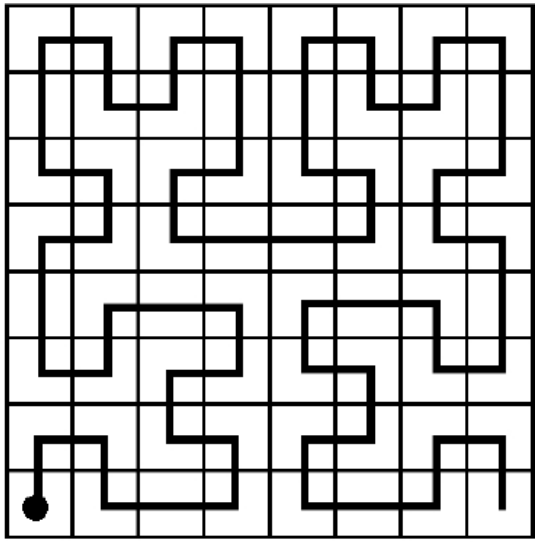
- **See lecture on** divide-&conquer Maxwell-Ehrenfest-surface hopping simulation

Razakh *et al.*, *PDSEC* (IEEE, '24); *PMBS* (IEEE, '24); *IEEE/ACM SC* ('25), doi: 10.1145/3712285.3771785

Computation Locality

Data to computation re-ordering: How to traverse cells?

- **Pair-interaction computation: Preserve nearest-neighbor cells' proximity in memory**
- **Spacefilling curve: Mapping from the *d*-dimensional space to one-dimensional list to preserve spatial proximity of consecutive list elements**



J. Mellor-Crummey *et al.*, *Int'l J. Parallel Prog.* **29**, 217 ('01)

Hilbert-Peano Curve

- **Gray code: a sequence of numbers such that successive numbers have Hamming distance 1**

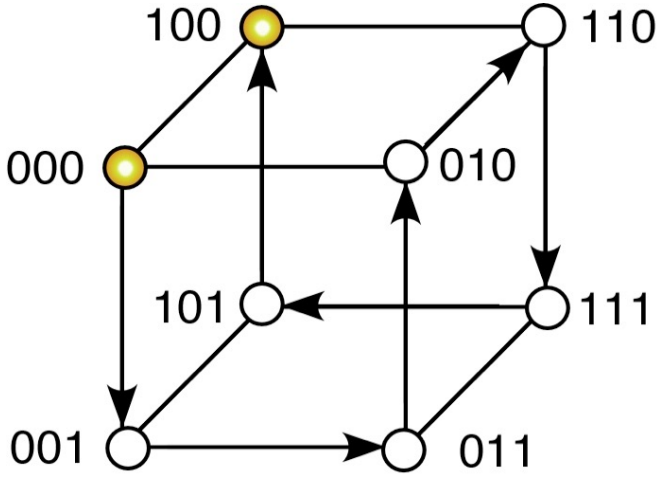
```
Algorithm: Recursive generation of k-bit Gray code G(k)
(1) G(1) is a sequence: 0 1.
(2) G(k+1) is constructed from G(k) as follows:
    a. Construct a new sequence by appending a 0 to the left of all members
       of G(k).
    b. Construct a new sequence by reversing G(k) & then appending a 1 to
       the left of all members of the sequence.
    c. G(k+1) is the concatenation of the sequences defined in steps a & b.
```

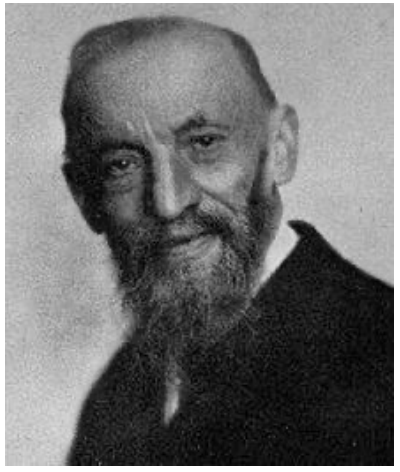
of bits where two binary numbers differ

- **G(3): 000 001 011 010 110 111 101 100**



David Hilbert (1862–1943)

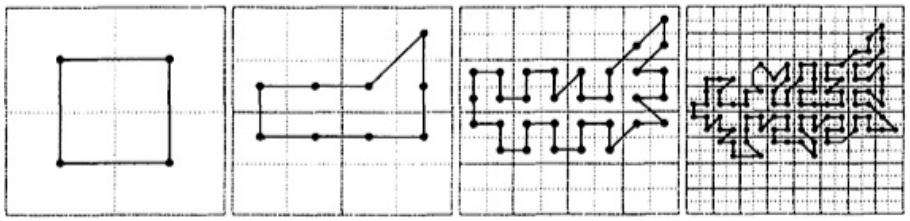




Giuseppe Peano (1858–1932)

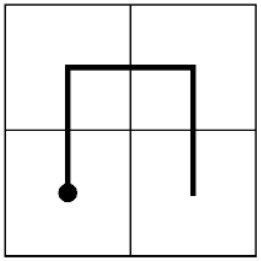
Hilbert-Peano Curve

- **Hilbert curve: recursive application of the *d*-dimensional Gray codes**
- **2-dimensional Hilbert curve**

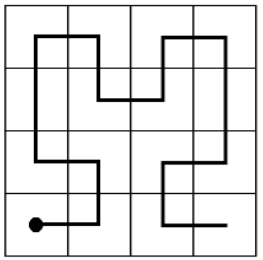


NP-complete traveling salesman problem

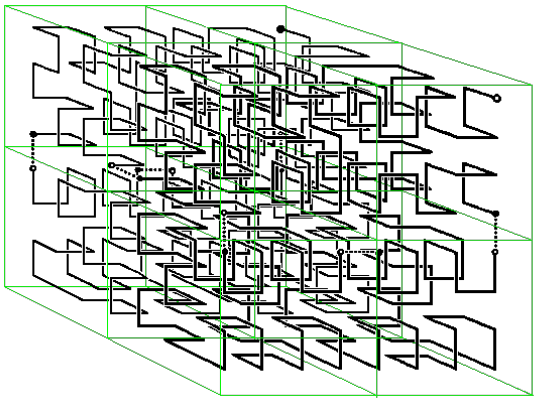
level 1



level 2

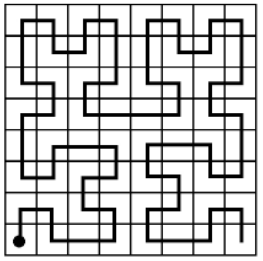


- **3-dimensional Hilbert curve**



See Corcoran *et al.*, *SC17*

level 3

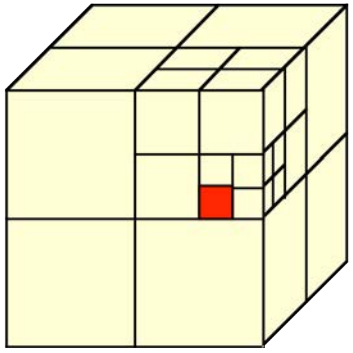


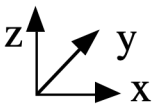
Morton (Z) Curve

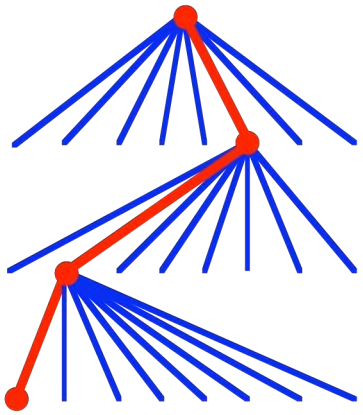
- **Spacefilling curve based on octree index**

x =	1	1	0
y =	0	0	0
z =	1	0	0
R =	101	001	000

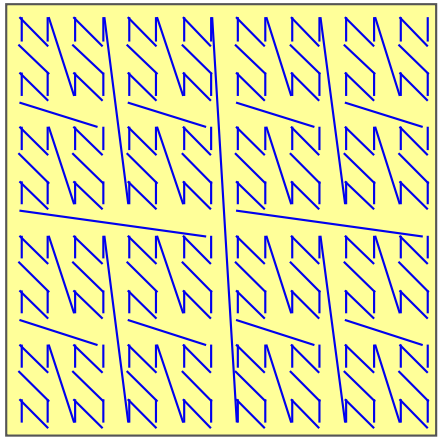
- **3D → list map preserves spatial proximity**
- **Multiresolution analysis made easy**

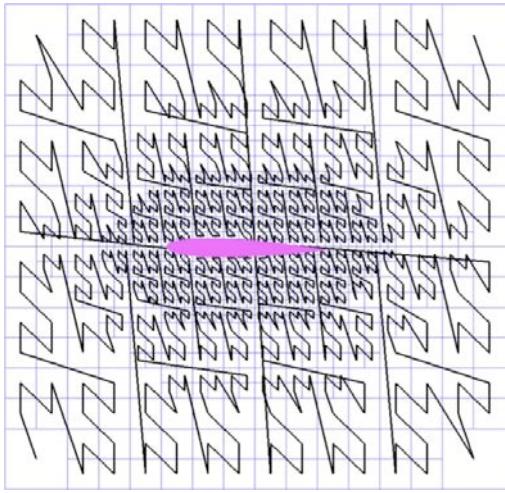






A. Omeltchenko *et al.*, *Comput. Phys. Commun.* **131**, 78 ('00)





Multiresolution analysis

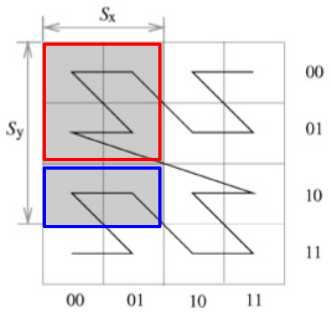
Analysis of Data Locaility

124 IEEE TRANSACTIONS ON KNOWLEDGE AND DATA ENGINEERING, VOL. 13, NO. 1, JANUARY/FEBRUARY 2001

Analysis of the Clustering Properties of the Hilbert Space-Filling Curve

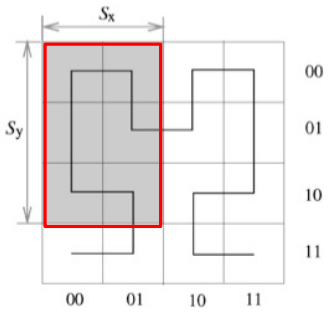
Bongki Moon, H.V. Jagadish, Christos Faloutsos, *Member*, *IEEE*, and Joel H. Saltz, *Member*, *IEEE*

Morton

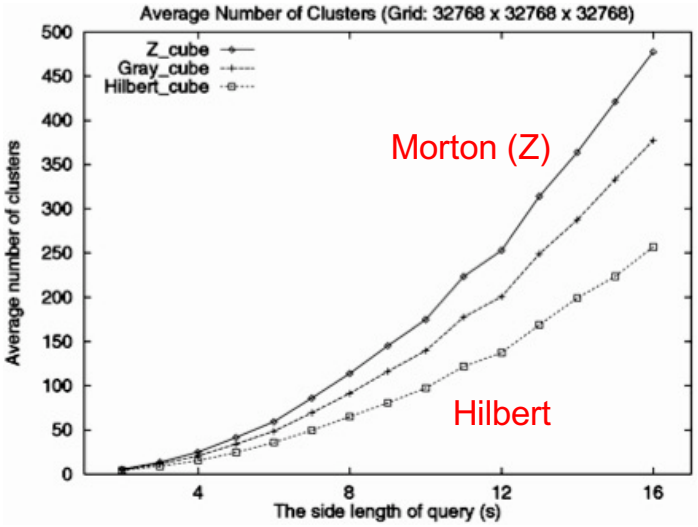


2 clusters

Hilbert



1 cluster



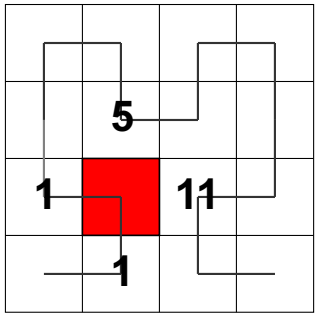
Cluster ~ cache line ~ latency cost

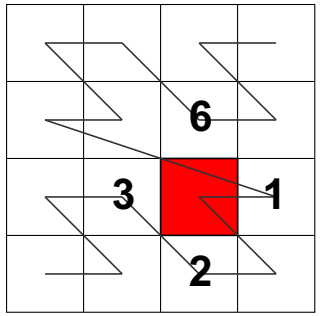
Hilbert curve is better than Morton curve for spatial range query

Alternative Locality Measure for MD

Which curve is better for spatial "pair" query?

- **Evaluate curves based on curve distances to neighbors**
- **Compare number below & above threshold cutoff k_c (like cache size)**





- 4x4 Hilbert:
  - 30 1s
  - 10 3s
  - 4 5s
  - 2 11s
  - 2 13s
- Lower median, higher variance
- Better for $k_c = 1$

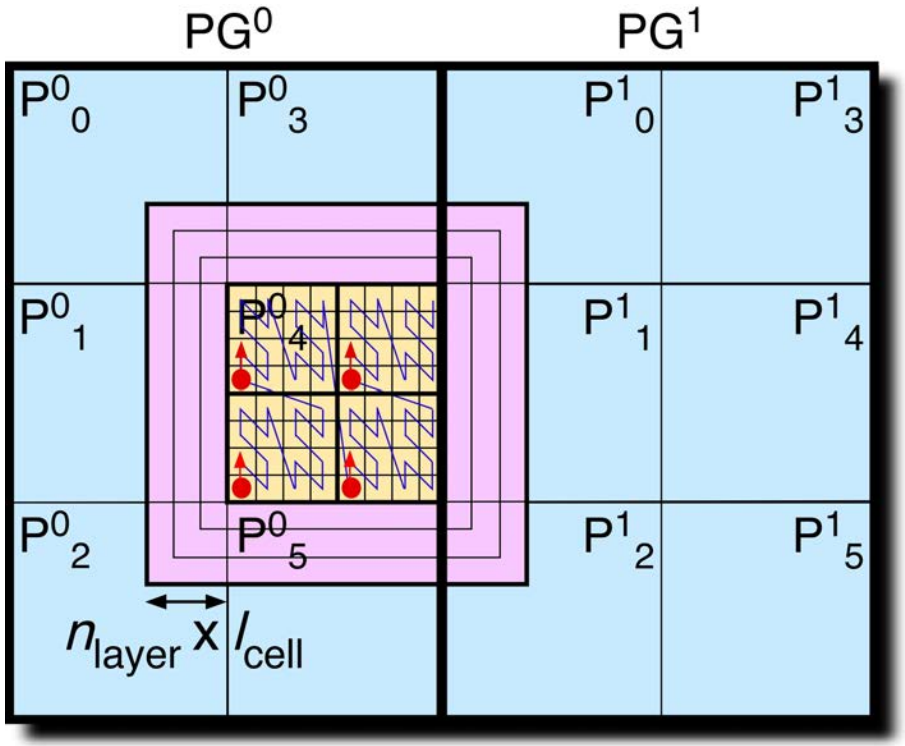
- 4x4 Z-curve:
  - 16 1s
  - 16 2s
  - 8 3s
  - 8 6s
- Higher median, lower variance
- Better for $2 < k_c < 13$

Scott Calaghan (CSCI 596 final project)

Tunable Hierarchical Cellular Decomposition

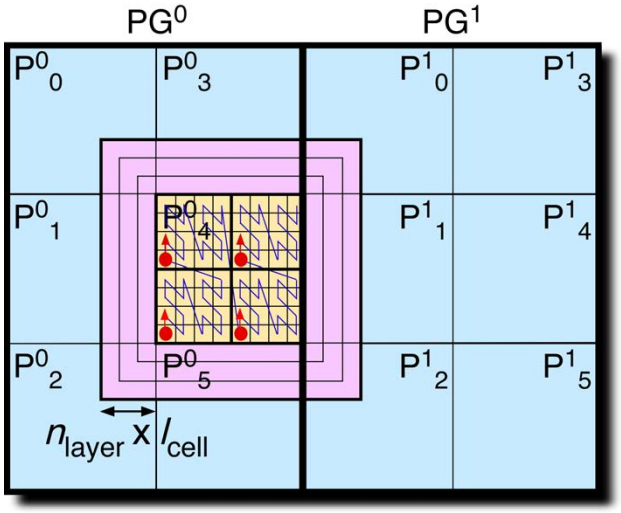
Mapping $O(N)$ divide-&-conquer algorithms onto memory hierarchies

- **Spatial decomposition with data "caching" & "migration"**
- **Computational cells (*e.g.*, linked-list cells in MD) < cell blocks (threads) < processes (P^{γ}_{π}, spatial decomposition subsystems) < process groups (P^{γ}, Grid nodes)**
- **Multilayer cellular decomposition (MCD) for n-tuples (n = 2–6)**
- **Tunable cell data & computation structures: Data/computation re-ordering & granularity parameterized at each decomposition level**
- **Tunable hybrid MPI + OpenMP + SIMD implementation**



Nomura *et al.*, *IPDPS 2009*

Performance Tunability

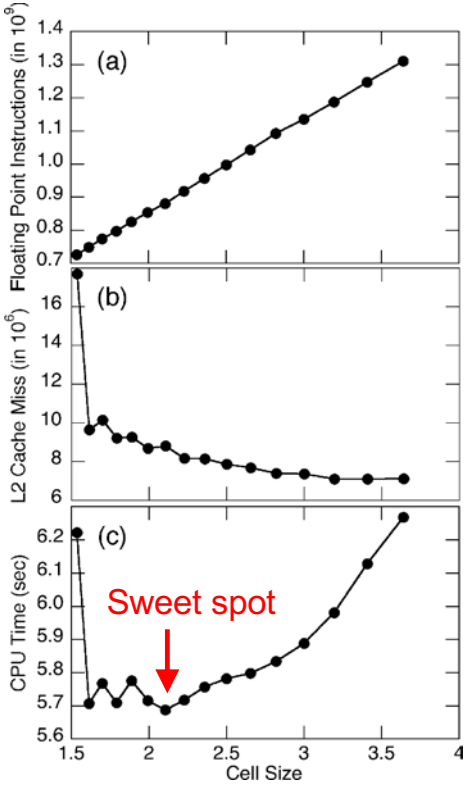


MPI/OpenMP parallelism trade-off:

8,232,000-atom silica MRMD & 290,304-atom RDX F-ReaxFF on 8-way 1.5 GHz Power4

Floating-point operation/L2 cache miss trade-off:

331,776-atom silica MRMD on 1.4 GHz Pentium III



Number of OpenMP threads, n_{td}	Number of MPI processes, n_p	Execution time/MD time step (sec)	
		MRMD	P-ReaxFF
1	8	4.19	62.5
2	4	5.75	58.9
4	2	8.60	54.9
8	1	12.5	120

SIMD Vectorization

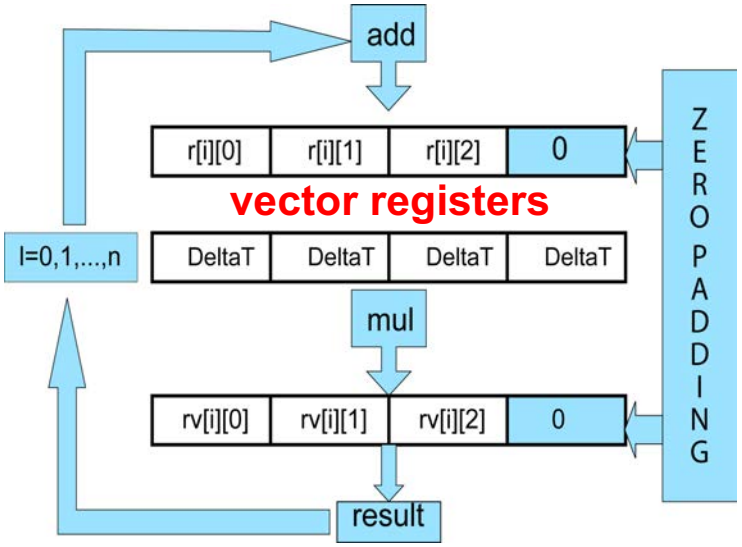
- **Single-instruction multiple-data (SIMD) parallelism**

(Example) Zero padding to align complex data

Original solution

```
for (i=0; i<N; i++)
  for (a=0; a<3; a++)
    r[i][a] =
    r[i][a] +
    DeltaT*rv[i][a];
```

SIMD solution

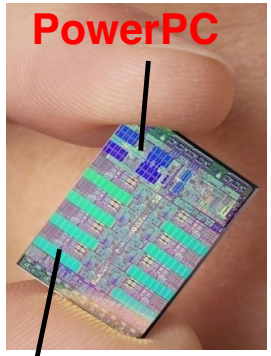


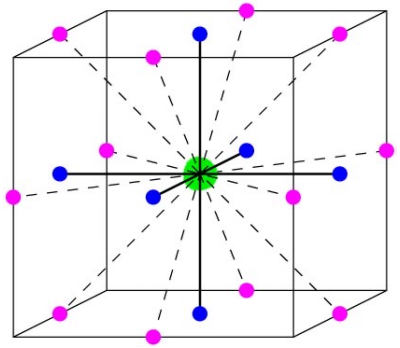
***cf.* False-sharing avoidance**

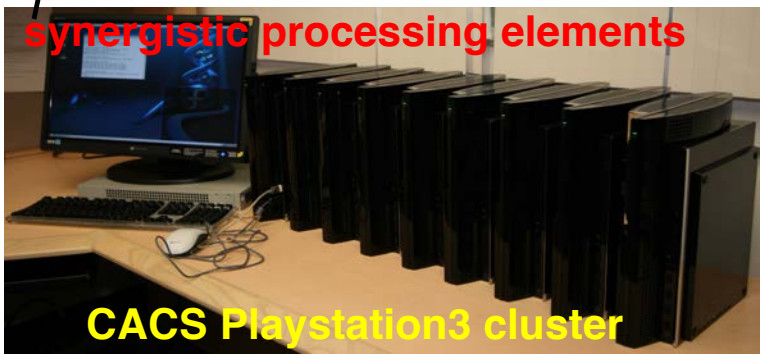
Peng *et al.*, *PDPTA 2009*; *UCHPC 2010; J. Supercomputing* **57**, 20 ('11)

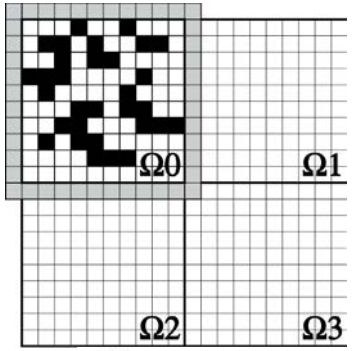
Hierarchical Parallelization

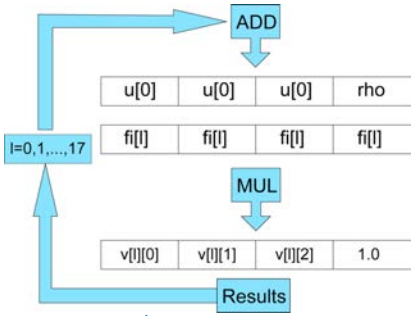
- **Developed a hierarchical parallel lattice Boltzmann method (pLBM) for flow simulation on a cluster of Cell Broadband Engine-based Playstation3 consoles & IBM BlueGenes**
  1. **Spatial decomposition *via* message passing**
  2. **Multithreading through critical section-free, dual representation**
  3. **Single-instruction multiple data (SIMD) parallelism *via* new vector transforms**





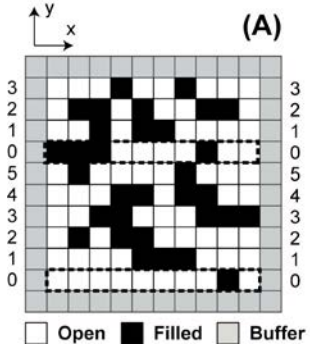


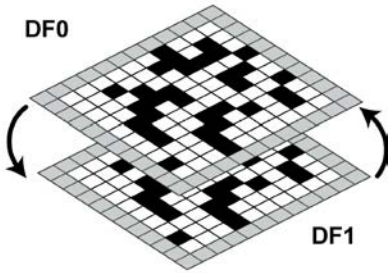




$\vec{j} = (\rho, j_x, j_y, j_z)$

(A) (B)





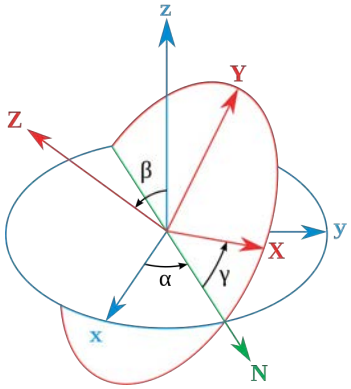
Peng *et al.*, *IJCS 08*; *Euro-Par 08*; *IPDPS 09*; *cf.* https://en.wikipedia.org/wiki/Four-vector

More Four-Vectors for SIMD

Use SIMD-efficient four-vectors abundant in mathematical physics!

- **Special relativity in physics: space (x, y, z)-time (t) four-vector**
  $X^{\mu} = (ct, x, y, z)$; c: light speed
- **Quaternion representation of rotation in computer graphics:**

$$\begin{bmatrix} q_0 \\ q_1 \\ q_2 \\ q_3 \end{bmatrix} = \begin{bmatrix} \cos\frac{\theta}{2}\cos\frac{\phi+\psi}{2} \\ \sin\frac{\theta}{2}\cos\frac{\phi-\psi}{2} \\ \sin\frac{\theta}{2}\sin\frac{\phi-\psi}{2} \\ \cos\frac{\theta}{2}\sin\frac{\phi+\psi}{2} \end{bmatrix}; \ (\theta, \phi, \psi)\text{: Euler angles}$$



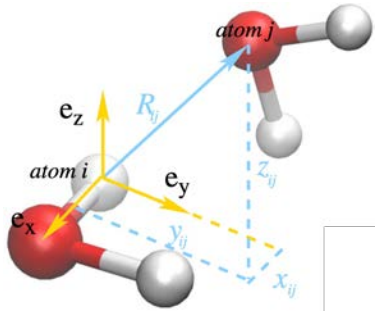
- **Feature vector in deep-learning molecular dynamics:**
  $D_{ij} = \left(1/R_{ij}, x_{ij}/R_{ij}, y_{ij}/R_{ij}, z_{ij}/R_{ij}\right)$

L. Zhang *et al.*, *Phys. Rev. Lett.* **120**, 143001 ('18)

2020 Gordon Bell Prize
Co-founder, DP Technology
Dean, AI for Science Institute

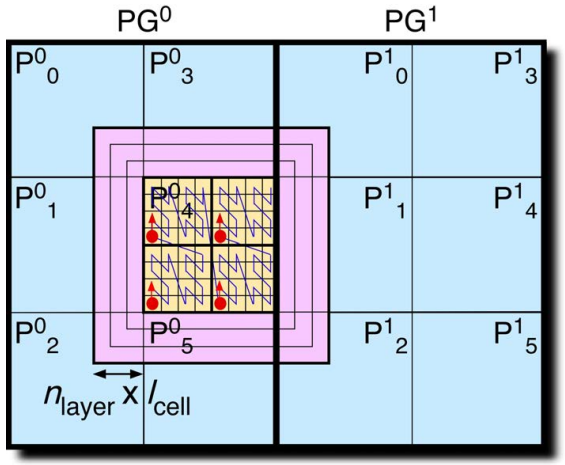
AI-for-Science start-up DP Technology raises US$114 million in Series C round

South China Morning Post
December 24, 2025 • 3 min read



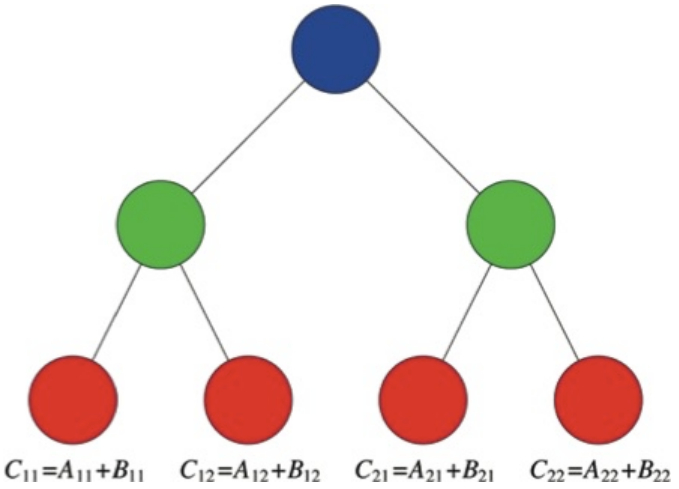
Recap: Intranode Performance Optimization

- **Key hardware feature:** Memory hierarchy; latency *vs*. bandwidth (*cf*. Internet speed test)
- **Key strategy:** Feed the fast processor by (1) enhancing data/computation locality to achieve high cache-hit rate (*e.g*., space-filling curve) & (2) increasing operational intensity, q = fast computation/slow memory I/O (*e.g*., BLAS-ify)
- **Do it yourself:** Hierarchical decomposition *via* divide-&-conquer implemented with hybrid MPI + OpenMP + SIMD (*e.g*., CUDA or OpenMP)



Cache-Oblivious Linked-List Cell MD?

- **Recursive blocking for cells?**



Cache-Oblivious Algorithms

EXTENDED ABSTRACT SUBMITTED FOR PUBLICATION. In *Proc. FOCS99*

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{athena,cel,prokop,sridhar}@supertech.lcs.mit.edu

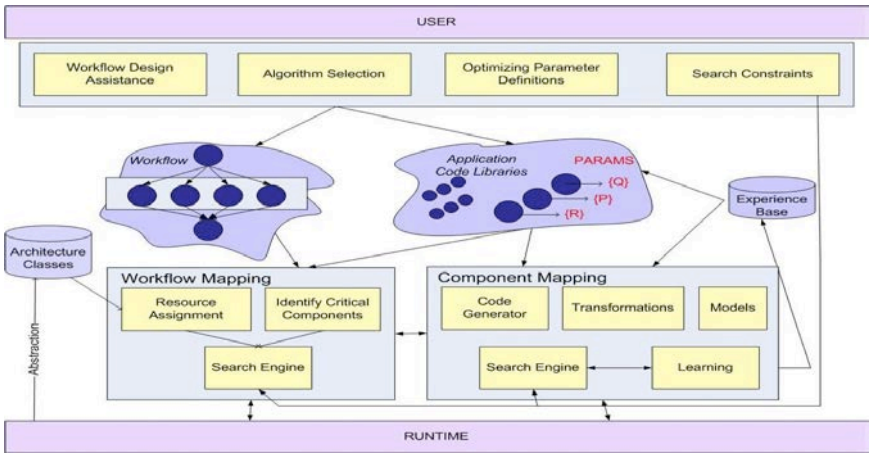
We introduce an "ideal-cache" model to analyze our algorithms, and we prove that an optimal cache-oblivious algorithm designed for two levels of memory is also optimal for multiple levels.

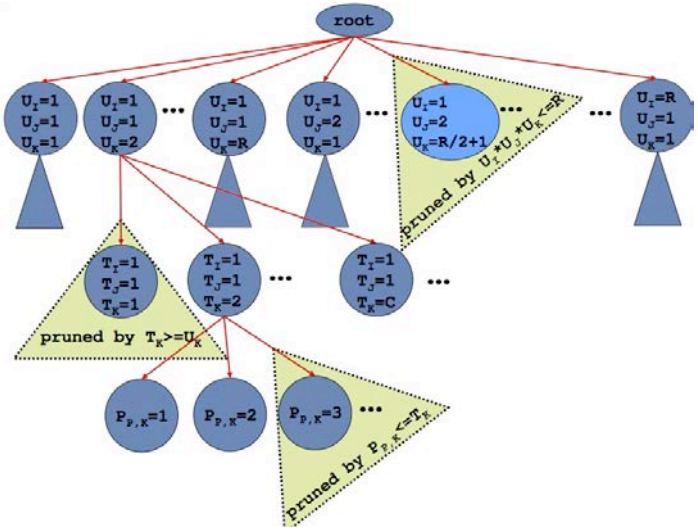
Intelligent Performance Optimization

Use AI!

- **Knowledge representation to *express* concurrency/data locality & machine learning to *optimally map* them to hardware**

  ***e.g.*, Tunable hierarchical cellular decomposition *exposes* maximal data locality**



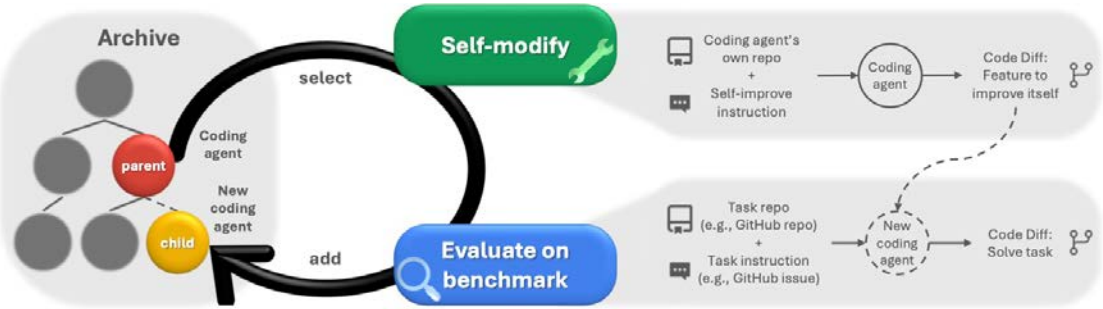


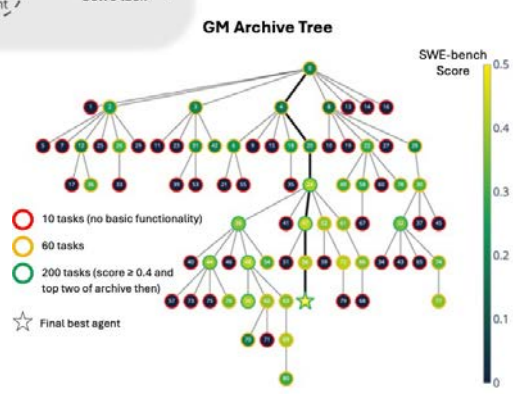
Pruned decision tree
C. Chen, *Ph.D. Thesis* (Computer Science, USC, '07)

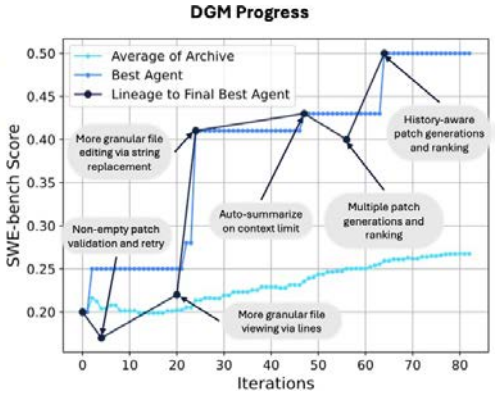
"Intelligent optimization of parallel & distributed applications," B. Bansal, U. Catalyurek, J. Chame, C. Chen, E. Deelman, Y. Gil, M. Hall, V. Kumar, T. Kurc, K. Lerman, A. Nakano, Y. L. Nelson, J. Saltz, A. Sharma, and P. Vashishta, in *Proc. of Next Generation Software Workshop, Int'l Parallel & Distributed Processing Symp. (IPDPS 07)*

Optimization by Self-Improving AI

- **Self-improving AI repeatedly modifies its own code**
- **Darwin Gödel model grows a tree-structured archive of generated high-quality coding agents**
  ***cf.* genetic programming** John Koza, https://mitpress.mit.edu/9780262111706/genetic-programming

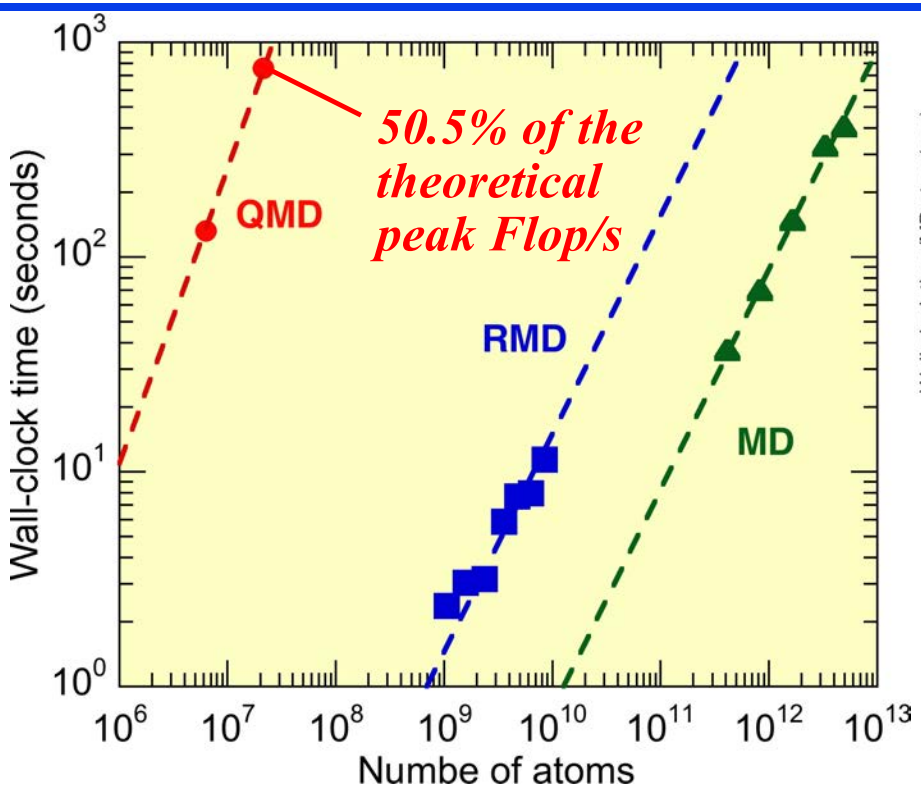


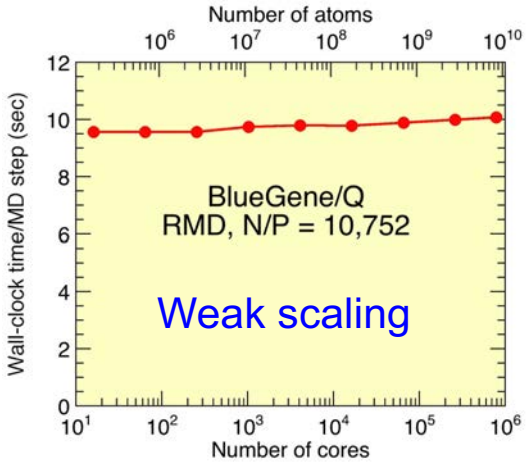




"Darwin Gödel machine: open-ended evolution of self-improving agents," J. Zhang *et al.*, *ICLR 2026*
https://arxiv.org/abs/2505.22954

Scalable Simulation Algorithm Suite





QMD (quantum molecular dynamics): DC-DFT

RMD (reactive molecular dynamics): F-ReaxFF

MD (molecular dynamics): MRMD

Nomura *et al.*, *IEEE/ACM SC14*

- **4.9 trillion-atom space-time multiresolution MD (MRMD) of SiO_2**
- **8.5 billion-atom fast reactive force-field (F-ReaxFF) RMD of RDX**
- **39.8 trillion grid points (50.3 million-atom) DC-DFT QMD of SiC**

Parallel efficiency 0.984 on 786,432 Blue Gene/Q cores

Scalability on Multicore Clusters

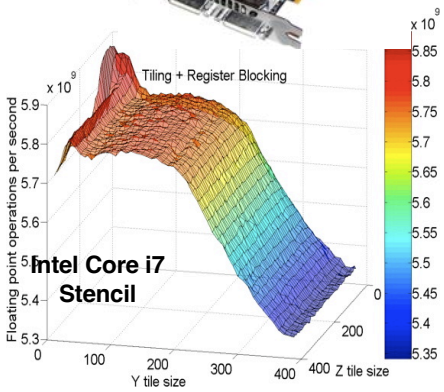
Hybrid message-passing (MPI) + multithreading (OpenMP) + single-instruction multiple-data (SIMD) programming

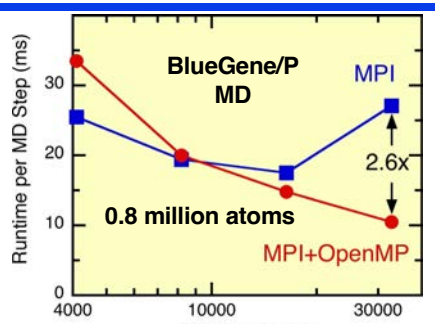
- 2.6× speedup over MPI by hybrid MPI+OpenMP on 32,768 IBM BlueGene/P cores
- Multithreading parallel efficiency 0.99 for MD on 64-core Godson-T processor
- SIMD efficiency 0.93 on PlayStation3
- 8.8× speedup on an NVIDIA GeForce 8800 GTS graphics processing unit (GPU) over an AMD Sempron CPU
- 55% of theoretical peak performance on 2.67 GHz Intel Core i7 920

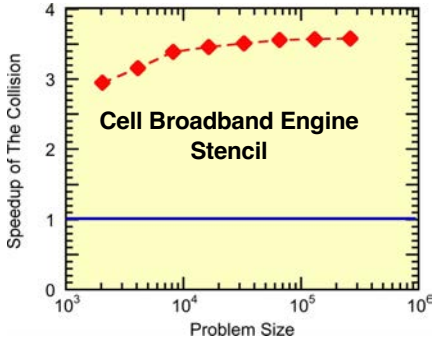


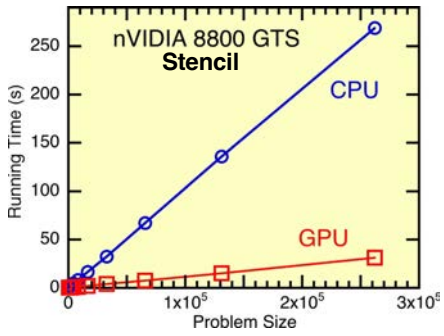












IJCS08; IPDPS09; PPL09; ICS10; CF11; Euro-Par12; JSC12; JPDC13; SC13

Multi-Exaflop/s NAQMD & NNQMD

- ***Achieving multi-Exaflop/s*: On 60,000 GPUs of Aurora: 1.87 Exaflop/s; 152× & 3,780× improvements of time-to-solution over state-of-the-art for 15.4M-electron NAQMD & 1.2T-atom NNQMD; nearly perfect parallel efficiency** **Razakh *et al.*, *2025 Gordon Bell prize finalist*** doi: 10.1145/3712285.3771785
- **Simulated light-matter dynamics in topological quantum materials: Petahertz (10^6× faster) & attojoule (orders-of-magnitude lower-power) electronics for sustainable AI society**
  cf. J. Zhao *et al*., *Science* **392**, 771 ('26)



Image: J. Insley, V. Mateevisti, S. Rizzi, J. Knowles

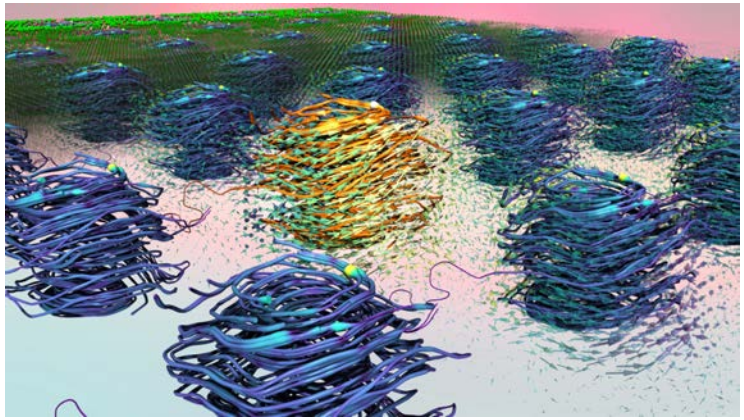


Photo-switching of skyrmion superlattice
https://www.youtube.com/watch?v=79Rp0v9cc4E

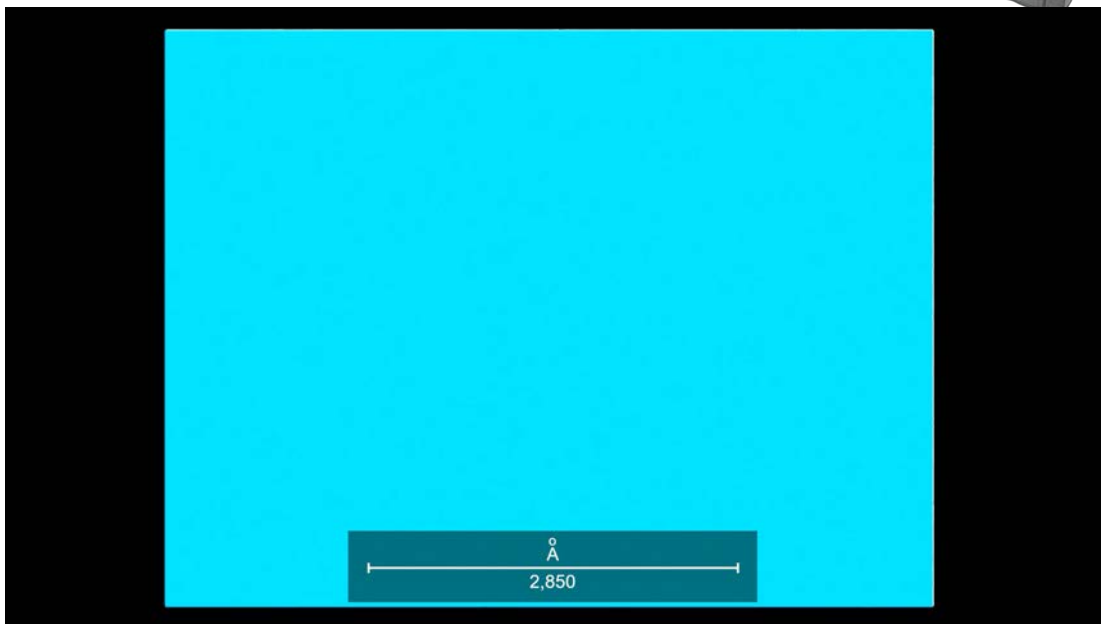
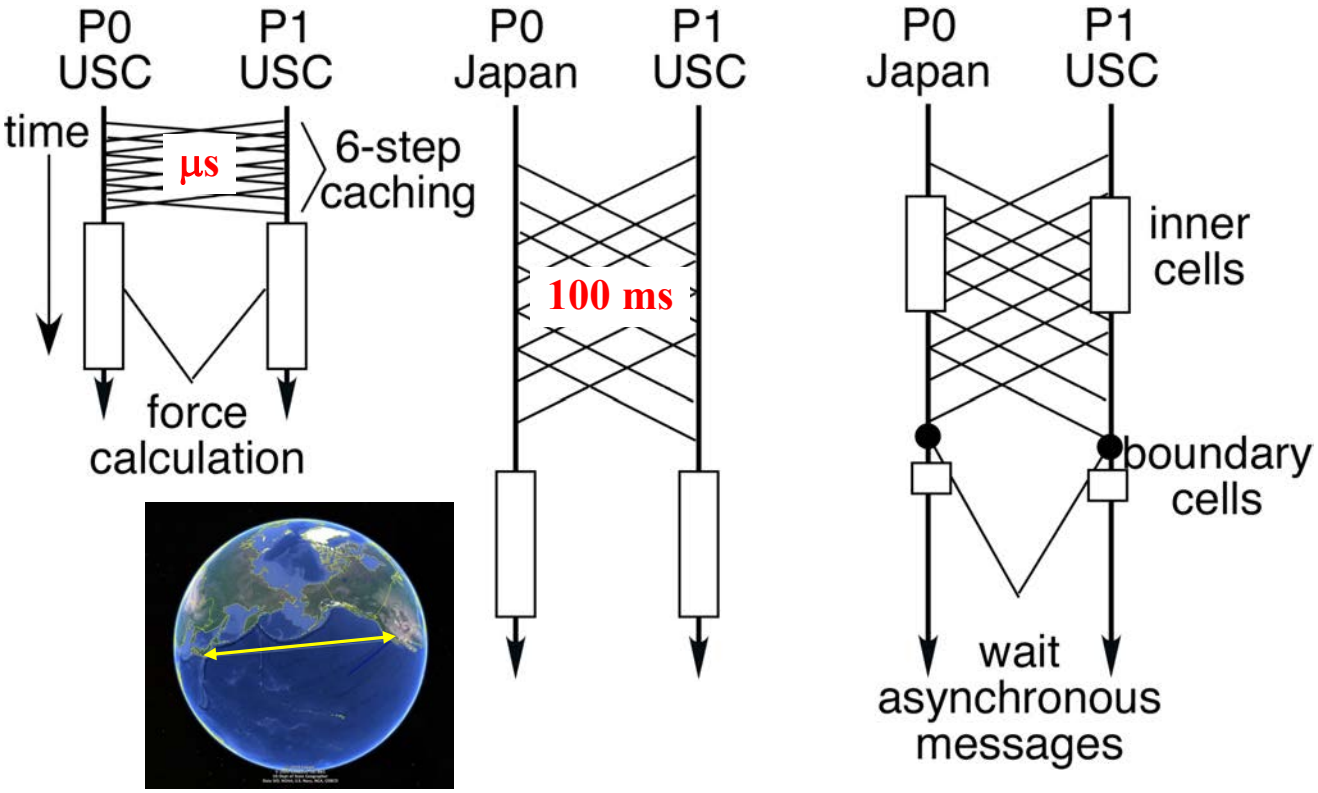


Photo-induced topological phase-transition dynamics
cf. Kibble-Zurek mechanism in cosmology

Internode Optimization

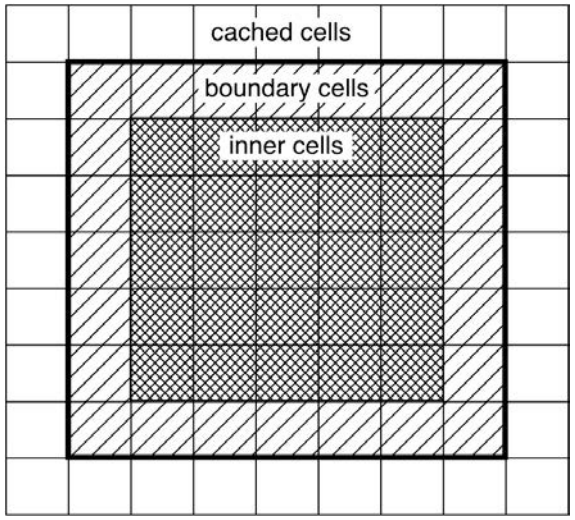
- **Communication bottleneck in metacomputing on a Grid**



Grid-Enabled MD Algorithm

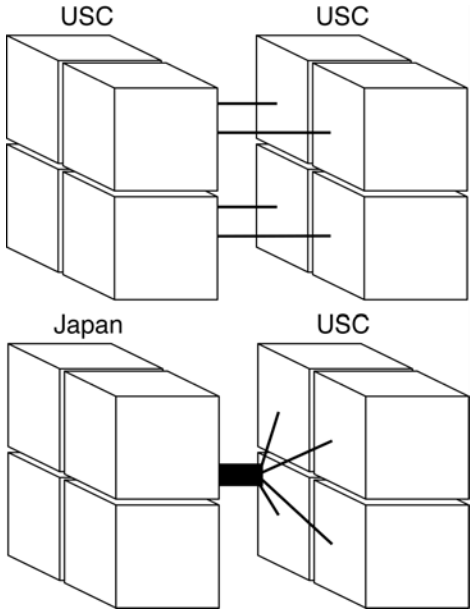
Grid MD algorithm:

1. asynchronous receive of cells to be cached MPI_Irecv()
2. send atomic coordinates in the boundary cells
3. compute forces for atoms in the inner cells
4. wait for the completion of the asynchronous receive MPI_Wait()
5. compute forces for atoms in the boundary cells



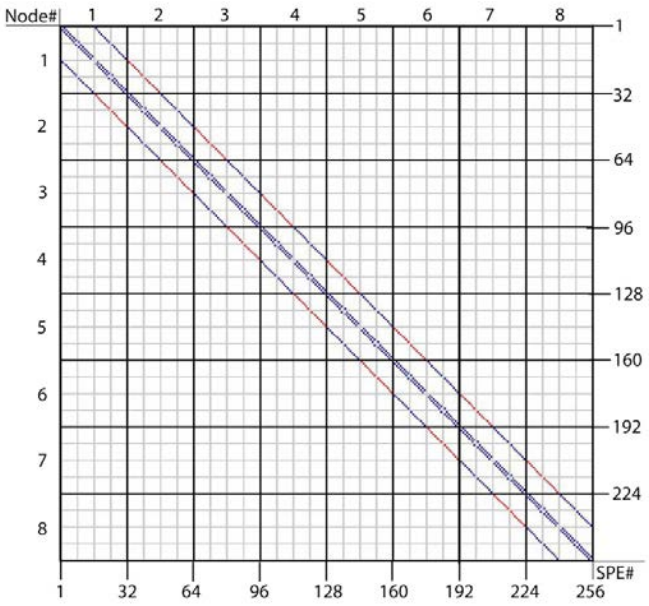
Renormalized Messages:

Latency can be reduced by composing a large cross-site message instead of sending all processor-to-processor messages

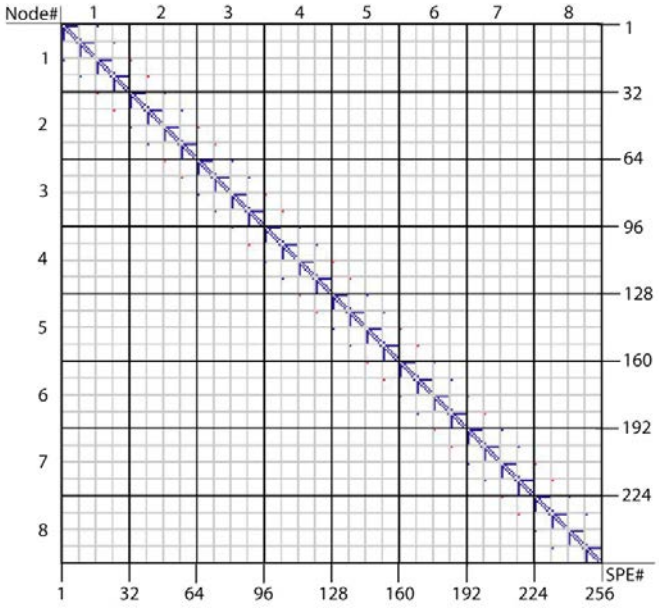


Renormalized Messages

Communication pattern of a 3D particle transport simulation code on a cluster of quad-Cell (32 cores) nodes*



Original



Renormalized

*LANL Roadrunner: first Petaflop/s computer

H. Dursun *et al.*, *Parallel Processing Letters* **19**, 535 ('09)

Where to Go from Here

- **Performance profiling: First thing to find is how well/badly your program is performing in terms of flop/s performance, vectorization, cache miss, *etc*.**
- **Use professional tools like Intel VTune & Advisor if available on your computer:**
  https://www.intel.com/content/www/us/en/developer/tools/oneapi/vtune-profiler.html
  https://www.intel.com/content/www/us/en/developer/tools/oneapi/advisor.html
  https://www.youtube.com/watch?reload=9&v=ymy139CuAx8

Advisor can show you the "roofline" of your application

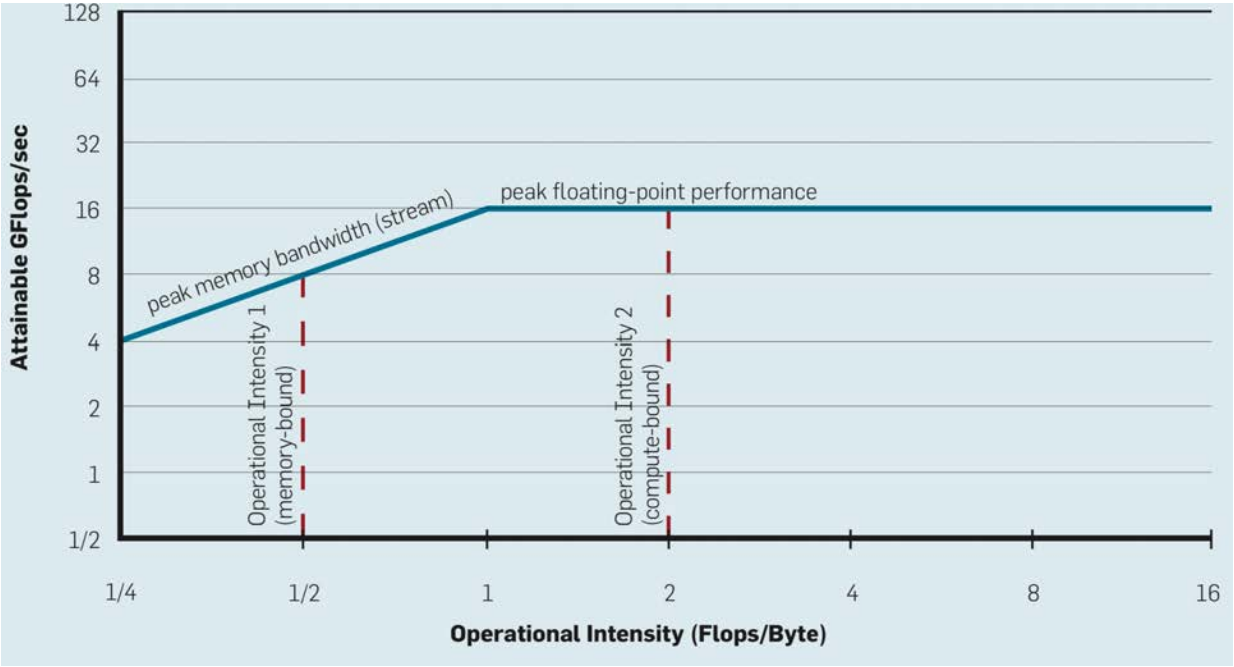
- **Off-chip memory bandwidth (from DRAM) is critical for performance (to feed enough data to be operated)**
- ***Operational intensity*: Operations per byte of DRAM traffic**
- ***Roofline model*: Predicts the floating-point (fp) performance from operation intensity, theoretical peak fp performance & peak memory bandwidth**

$$Attainable fp\ performance\ \left[\frac{\text{Gflop}}{\text{sec}}\right] = \min\left(\begin{array}{r} Peak\ fp\ performance\ \left[\frac{\text{Gflop}}{\text{sec}}\right], \text{ cf. assignment 1} \\ Peak\ memory\ bandwidth\ \left[\frac{\text{GByte}}{\text{sec}}\right] \times Operational\ intensity\ \left[\frac{\text{flop}}{\text{Byte}}\right] \end{array}\right)$$

S. Williams *et al*., *Commun. ACM* **52(4)**, 65 ('09)
V. Elango *et al*., *ACM. T. Arch. Code Opt.* **11**, 67 ('15)

Roofline Model of Performance

$$Attainable\ fp\left[\frac{\text{Gflop}}{\text{sec}}\right] = \min\left(Peak\ fp\left[\frac{\text{Gflop}}{\text{sec}}\right], Memory\ bandwidth\left[\frac{\text{GByte}}{\text{sec}}\right] \times Operational\ intensity\ \left[\frac{\text{flop}}{\text{Byte}}\right]\right)$$



Key: Data/computation locality

see Berkeley CS267 lecture on “memory hierarchies & matrix multiplication”